TEST PLAN
Multi-Modal Agentic AI System for Medical Diagnosis and Triage

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GitHub: https://github.com/abhishek7112000/Multi-Modal-Agentic-AI-System-for-Medical-Diagnosis-and-Triage

1. Purpose and Scope

This plan verifies the system against Pipeline Workflow v2.0 — every stage checked in isolation and end-to-end, all KPIs evidenced, and clinical safety protected.
Pipeline: **data pre-processing → model training → evaluation → ONNX export → agent deployment.**

Pipeline stage	What we are testing	Test type
Data pre-processing & feature engineering	Loaders, splits, image/tabular/text transforms for all 3 datasets; zero leakage confirmed.	Data validation, unit
Model training & tuning (M1–M3)	Training loops complete; hyperparameter search reproducible with fixed seeds.	Unit, integration
ONNX export & quantisation	PyTorch ↔ ONNX parity; quantised sizes within Vercel 250 MB limit.	Unit, integration
Model evaluation	Held-out test performance meets every agreed KPI threshold.	Model performance
LangGraph agent + MedGemma + RAG	Node routing correct; DiagnosisReport JSON valid; RAG chunks relevant.	Unit, integration, safety
Next.js app + Calendar/Gmail	API routes correct; booking end-to-end; first-time user navigates unaided.	Integration, UAT

Out of scope: clinical certification, real patient data, third-party weight internals (MedGemma, OpenMed base), production load/penetration testing. MedGemma-27B-IT is verified for output validity and safety only — not trained or evaluated as one of the three ML models.

2. Coding Standard

One standard across both languages enforced in CI. No PR is merged until every automated check is green — no exceptions.

Area	Python — model training	TypeScript — app and agent
Format & style	Black + isort; PEP 8 baseline.	Prettier + ESLint; enforced on save.
Lint & types	Flake8/Pylint; mypy strict — no untyped signatures.	TypeScript strict mode; ESLint rules set to error.
Naming	snake_case functions/variables; PascalCase classes.	camelCase functions/variables; PascalCase components/types.
Documentation	Google-style docstrings on all public functions.	JSDoc/TSDoc on exports; README per module and at root.
Errors & logging	Specific exceptions; logging module; no secrets or PII in logs.	Typed errors; ONNX fallback if upstream fails; no keys logged.
Version control	Protected main; feature branches; one sign-off per PR; imperative commits — both languages.	Same branch protection and PR process; shared repo and commit convention.

3. Data Quality Testing and Validation

Data checks run before any training and are part of CI — a bad preprocessing change blocks a merge exactly as a failing unit test would.
NIH ChestX-ray14 (M1): All 112,120 files open; each image 224×224 three-channel RGB, ImageNet-normalised; all Finding_Labels within 14 known classes; official patient-level split with zero ID overlap; class frequencies logged to confirm imbalance handling needed.

Synthea EHR (M2): Six CSVs validated for schema/types; filtered to ICD-10 I21, I25, I50, I48, J18, J44, J45; vitals/labs range-checked; group-median imputation with binary is_imputed column confirmed; duplicate patient IDs removed.
Medical QA Dataset (M3): Loaded from HuggingFace (Malikeh1375/medical-question-answering-datasets); all four subsets confirmed — MedQA, MedMCQA, PubMedQA, HealthSearchQA; Q→A pairs tokenised with OpenMed tokeniser at max_length=512; chest/cardiovascular pairs retained; no train/val/test overlap.
Encyclopedia/RAG: 512-token segments, 64-token overlap via LangChain RecursiveCharacterTextSplitter; ChromaDB embeddings non-zero; smoke-test queries return relevant chunks. **Cross-cutting:** Fixed seeds; stratified splits within 2%; no patient or question in more than one split.

4. Unit and Integration Testing

All functions tested in isolation; CI runs on every push; target ≥ 80% line coverage via pytest-cov and Istanbul. Integration test drives the full agent graph: **intake_node → route_node → xray_node/clinical_node → synthesise_node → clarify_node → rag_node → response_node → triage_node.**

Component / function	Example input	Tool	Pass criterion
preprocess_image()	Arbitrary-size JPEG.	pytest	224×224×3 tensor; values in ImageNet-normalised range.
Metric calc (AUC / F1 / ROUGE-L)	Synthetic inputs, known ground truth.	pytest	AUC = 1.0 on perfect predictor; ROUGE-L = 1.0 on identical strings.
ONNX parity — M1 and M2	100 records through PyTorch and ONNX.	pytest + onnxruntime	M1 $\|\Delta\| < 1e-4$; M2 ≤ 0.01; M3 ROUGE-L delta < 0.5 pp.
assign_triage()	Severity = 3; SpO2 < 90%.	jest	Returns EMERGENCY without exception.
Agent nodes (route / synthesise / clarify)	Mocked outputs; confidence = 0.55.	jest	Correct edge; clarify_node fires because 0.55 < 0.60 threshold.
API routes /api/models/*	Well-formed request payload.	jest	Valid JSON matching schema; status 200.
Full end-to-end flow	Synthetic symptoms + chest X-ray fixture.	Playwright	Complete DiagnosisReport returned; triage assigned; no errors.

5. Model Performance Evaluation Against KPIs

All three models are evaluated on the same held-out test set. Tuning uses Optuna (M1, M2 fusion) and GridSearchCV (XGBoost) on the validation set only; pre/post-tuning gains and an ablation (XGBoost-only / text-only / hybrid) are recorded. Each KPI passes only when its target is met.

Model	KPI targets	Test method
M1 — ChestVision (transfer learning; ResNet50V2 vs ViT-B/16 vs EfficientNetV2B0 — ViT-B/16 selected)	Mean AUC-ROC ≥ 0.82 (CheXNet benchmark 0.841) | Per-class AUC ≥ 0.75 | F1 ≥ 0.60 for top 5 conditions | Grad-CAM heatmaps validated vs NIH bounding-box annotations | ONNX CPU < 50 ms/image	NIH held-out test set; best arch by mean AUC across all 3 fine-tuned architectures; Grad-CAM scored vs annotated regions; latency via onnxruntime CPU.
M2 — ClinicalFusion (hybrid MLP tabular + frozen BioClinical-ModernBERT-	Macro F1 ≥ 0.80 | Weighted F1 ≥ 0.85 | ROC-AUC ≥ 0.90/class (one-vs-rest) | Cohen's Kappa ≥ 0.75 (severity) | Severity MAE < 0.40 | ONNX CPU < 20 ms/record | False-low rate (critical → low severity) strictly < 2% — breach blocks deployment regardless of all other metrics.	Held-out Synthea test set; ablation table and confusion matrices in report; misclassified critical patients logged to audit file and reviewed manually before sign-off.

Model	KPI targets	Test method
large) + Clinical safety (hard gate)		
M3 — OpenMed QA (pre-trained + transfer learning; HF Endpoint)	Entity F1 ≥ 0.82 (≥ 0.80/type); DISEASE recall ≥ 0.85; transfer gain ≥ 3 F1 pts vs base.	seqeval span-level eval.
ONNX deployment (M1 and M2)	Quantised size within Vercel 250MB limit; output parity within tolerance.	Size + parity checks.
Latency (all)	M1 < 50ms/img; M2 < 20ms/record; M3 < 30ms/note (CPU).	onnxruntime CPU benchmark.

6. Triage Safety and User Acceptance Testing

Four triage levels: EMERGENCY, URGENT, ROUTINE, SELF_CARE. EMERGENCY displays A&E guidance and sends a GP alert email — no appointment booked. Agent confidence below 0.60 triggers the clarify_node; patient answers follow-up questions before a final report is produced. The agent must never downgrade a serious finding. All six UAT scenarios must pass before Austin (Team Leader) signs off.

ID	Scenario	Acceptance criterion
UAT01	Standard journey — X-ray uploaded and symptoms entered.	Diagnosis, severity, and triage level returned within target time; all fields populated; no errors shown.
UAT02	Cardiac-suggestive input — elevated troponin, chest pain, prior MI.	Cardiology referral at correct urgency; Google Calendar appointment booked; Gmail confirmation sent.
UAT03	Emergency presentation — SpO2 < 90%, severe chest pain.	EMERGENCY assigned; A&E guidance displayed; GP alert email sent; no appointment booked.
UAT04	User uploads unsupported file format (.txt or .bmp).	Friendly error shown; app does not crash; user can retry without refreshing.
UAT05	Brand-new user navigating without any help.	User completes full session from symptom entry to triage result without needing guidance.
UAT06	Agent confidence = 0.55 (below 0.60 threshold).	clarify_node fires; follow-up questions shown; DiagnosisReport produced only after patient responds.

7. Code Review and Sign-off Process

Workflow: **feature branch → pull request → CI green → peer review → merge.** Nothing reaches main without automated checks passing and a human reviewer approving. Peer review checklist: matches Pipeline Workflow spec; coding standard followed; tests pass; no secrets committed; ONNX parity re-validated for model changes. All four gates must pass in order.

Gate	Criteria to pass	Sign-off authority
Automated CI	All checks green — formatting, linting, type checking, unit tests, ONNX parity. A single failure blocks the merge. No overrides.	Enforced by GitHub Actions.
Peer code review	All five checklist items satisfied; every reviewer comment resolved; approval recorded in the PR.	Victor — Code Review Manager
Test sign-off	Data validation, unit and integration tests green; all KPIs from Section 5 met or exceeded; results documented.	Karan — Test Plan Manager
Release / UAT sign-off	All six UAT scenarios UAT01–UAT06 passed. Any failure requires a fix and full re-test before sign-off can be re-requested.	Austin — Team Leader